

CE 554: Prestressed Concrete Structures

Course Code	CE 554
Course Name	Prestressed Concrete Structures
L-T-P-C	3-0-0-3
Prerequisites	(i) Strength of Materials and Structures (CE301 or equivalent), (ii) Design of Reinforced Concrete Structures (CE351 or equivalent)
Intended for	UG (4th year), PG
Distribution	Discipline Elective (UG) / Elective (PG)
Approval	16th BoA (Feb 2017); 15th Senate (Feb 2018)

Course Contents

- **Introduction:** History of development, material and basic concept of prestressing, System of prestressing, Need and types of prestressing, Codes and Standards. [3 Lectures]
- **Analysis of prestressed members:** Basic assumptions, Analysis procedure, Prestress line and internal resisting couple, Concept of load balancing, Stresses in tendons, De-bonding and draping of prestressing tendons, Camber of prestressed member, Indeterminate prestressed concrete structures, Circular prestressing. [6 Lectures]
- **Losses of prestress:** Nature of losses of prestress, Consideration of allowable losses in design. [3 Lectures]
- **Deflection of prestressed concrete members:** Importance of deflection control, Influencing factors, Short term and long term deflection, Deflection of cracked members. [6 Lectures]
- **Prestressed concrete members under flexure, shear and torsion:** Flexural failure and strength analysis, Strain compatibility, Shear and torsion in prestressed members. [7 Lectures]
- **Module VI:** Stress transfer in pretensioned members, Anchorage zone stress in post-tension members, Bursting/ end-block stresses, Transmission and transfer lengths. [3 Lectures]
- **Design of prestressed concrete section:** Limit state design criteria for prestressed concrete members, Design of section for flexure, Design of section for axial tension, Design of section for compression, Design of section for shear and torsion, Design for bond and bearing. [8 Lectures]
- **Design of members:** Design of pretensioned and post-tensioned flexural member, Design of prestressed beam and slab, Partially prestressed member. [6 Lectures]

Text Books

1. N.K. Raju, restressed Concrete, 5th Edition, McGraw Hill Education, 2012.
2. P. Dayaratnam, Prestressed Concrete Structures, 5th Edition, Oxford & IBH Publishing Co. Pvt. Ltd., 2016.

References

1. Michael P. Collins and Denis Mitchell, Prestressed Concrete Structures, Prentice Hall, 1994.
2. N. Rajagopalan, Prestressed Concrete, 2nd Edition, Narosa Publishing House, 1994.
3. T. Y. Lin, N. H. Bums, Design of Prestressed Concrete Structures, 3rd Edition, Wiley India Private Limited, 2010.
4. A. E. Naaman, Prestressed Concrete Analysis and Design - Fundamentals, McGraw Hill Education, 2013.
5. E. G. Nawy, Prestressed Concrete - A Fundamental Approach, 5th Edition, Prentice Hall International, 2005.
6. IS 1343, Prestressed Concrete - Code of Practice, Bureau of Indian Standards, 2012.